# Fall 2023

**Problem 1 (Sp03, Fa23).** Prove that for each integer $n \geqslant 0$ there is a polynomial $T_n(x)$ with integer coefficients such that the identity

$$2\cos nz = T_n(2\cos z)$$

holds for all $z$.

**Problem 2 (Sp00, Fa23).** Let $\{f_n\}$ be a uniformly bounded, pointwise equicontinuous sequence of real valued functions on the compact metric space $(X, d)$. Define the functions $g_n : X \to \mathbb{R}$, for $n \in \mathbb{N}$ by

$$g_n(x) = \max\{f_1(x), \ldots, f_n(x)\}$$

Prove that the sequence $\{g_n\}$ converges uniformly.

**Problem 3 (Su84, Fa23).** Consider the solution curve $(x(t), y(t))$ to the equations

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 1 + \frac{1}{2}x^2 \sin y$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = 3 - x^2$$

with initial conditions $x(0) = 0$ and $y(0) = 0$. Prove that the solution must cross the line $x = 1$ in the $xy$ plane by the time $t = 2$.

**Problem 4 (Fa03, Fa23).** Show that the initial value problem

$$f''(z) = zf(z) \qquad f(0) = 1 \qquad f'(0) = 1$$

has an unique entire solution in the complex plane.

**Problem 5 (Sp03, Fa23).** Let $f$ be an entire function such that $\Re f(z) \geqslant -2$ for all $z \in \mathbb{C}$. Show that $f$ is constant.

**Problem 6 (Fa84, Fa23).** Let $\mathbb{R}[x_1, \ldots, x_n]$ be the polynomial ring over the real field $\mathbb{R}$ in the $n$ variables $x_1, \ldots, x_n$. Let the matrix $A$ be the $n \times n$ matrix whose i-th row is $(1, x_i, x_i^2, \ldots, x_i^{n-1})$, $i = 1, \ldots, n$. Show that

$$\det A = \prod_{i>j} (x_i - x_j)$$

**Problem 7 (Fa23).** Show that if $A$ and $B$ are unitary matrices of the same size, then all eigenvalues of $A + B$ have absolute value at most 2.

**Problem 8 (Fa23).** Given a finite group $G$, let $e(G)$ be the least common multiple of the orders of the group's elements. Prove that if the group is abelian, then it contains an element of order $e(G)$, and give a counterexample to this statement for non abelian groups.

**Problem 9 (Fa23).**

1. Give an example of an irreducible polynomial $f$ of degree 100 over the rationals $\mathbb{Q}$ such that $f$ splits into linear factors over the field $\mathbb{Q}[x]/\langle f(x)\rangle$.

2. Give an example of an irreducible polynomial $g$ of degree 100 over the rationals $\mathbb{Q}$ such that $g$ does not split into linear factors over the field $\mathbb{Q}[x]/\langle g(x)\rangle$.

**Problem 10 (Fa86, Fa13, Fa19, Fa23).** Show that the polynomial $p(z) = z^5 - 6z + 3$ has five distinct complex zeros, of which exactly three (and not five) are real.

**Problem 11 (Fa23).** Suppose that $f$ is a nondecreasing real valued function on $[0, 1]$. Find functions $g$ and $h_n$ such that

$$f(x) = g(x) + \sum_n h_n(x)$$

where $g$ is continuous, and each of the finite or countable number of functions $h_n$ has one point where it is not continuous, and is zero below this point and constant above it.

**Problem 12 (Su84, Fa23).** Let $\varphi(s)$ be a $\mathcal{C}^2$ function on $[1, 2]$ with $\varphi$ and $\varphi'$ vanishing at $s = 1, 2$. Prove that there is a constant $C > 0$ such that for any $\lambda > 1$,

$$\left|\int_1^2 e^{i\lambda x}\varphi(x)\,\mathrm{d}x\right| \leqslant \frac{C}{\lambda^2}$$

**Problem 13 (Fa86, Fa23).** Let the points $a$, $b$, and $c$ lie on the unit circle of the complex plane and satisfy $a + b + c = 0$. Prove that $a$, $b$, and $c$ form the vertices of an equilateral triangle.

**Problem 14 (Su84, Fa23).** Let $\rho > 0$. Show that for $n$ large enough, all the zeros of

$$f_n(z) = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots + \frac{1}{n!z^n}$$

lie in the circle $|z| < \rho$.

**Problem 15 (Su84, Fa23).** Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a real matrix with $a, b, c, d > 0$. Show that $A$ has an eigenvector $(x, y) \in \mathbb{R}^2$ with $x, y > 0$.

**Problem 16 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23).** Let $\mathbb{F}$ be a finite field with $q$ elements and let $V$ be an $n$-dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in $V$.

2. Let $\mathrm{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\mathrm{GL}_n(\mathbb{F})$.

3. Let $\mathrm{SL}_n(\mathbb{F})$ denote the subgroup of $\mathrm{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\mathrm{SL}_n(\mathbb{F})$.

**Problem 17 (Fa23).** If $H$ is a subgroup of a group $G$ of index 5, prove that $H$ contains a normal subgroup of $G$ of index at most 120. Give an example of a group $G$ and a subgroup $H$ of index 5 such that $H$ does not contain a normal subgroup of $G$ of index less than 120.

**Problem 18 (Fa23).** Show that the group $\mathbb{Q}$ of rational numbers under addition has no subgroups of finite index greater than 1.